



Advanced Time Series Analysis and Forecasting

Chapter 13: Foundation models and conformal prediction



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Acronyms used in this material (1/2)

- ACI: Adaptive Conformal Inference
- AI: Artificial Intelligence
- AIC: Akaike Information Criterion
- API: Application Programming Interface
- AR: AutoRegressive (model); in event studies: Abnormal Return
- ARX: AutoRegressive model with eXogenous variables
- BET: Bucharest Exchange Trading (index)
- BH: Benjamini–Hochberg (procedure)
- BOOM: Benchmark Of Observability Metrics
- CEE: Central and Eastern Europe
- CPM: Contiguous Patch Masking
- CPU: Central Processing Unit
- CQR: Conformalized Quantile Regression
- CRPS: Continuous Ranked Probability Score
- DM: Diebold–Mariano (test)
- DQ: Dynamic Quantile (test of Engle and Manganelli)
- ENTSO-E: European Network of Transmission System Operators for Electricity
- EODHD: EOD Historical Data (market-data provider)
- ETS: Error, Trend, Seasonality (exponential smoothing state space models)
- EU: European Union
- FDR: False Discovery Rate
- FWER: Family-Wise Error Rate
- GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- GIFT: General Time Series Forecasting Model Evaluation (the GIFT-Eval benchmark)
- GPT-2: Generative Pre-trained Transformer 2 (OpenAI language model)
- GPT4TS: GPT for Time Series (One Fits All)

Acronyms used in this material (2/2)

- GW: **Giacomini–White** (test of conditional predictive ability)
- HAC: **Heteroskedasticity and Autocorrelation Consistent** (standard errors)
- HAR: **Heterogeneous AutoRegressive** (model)
- HICP: **Harmonised Index of Consumer Prices**
- HLN: **Harvey–Leybourne–Newbold** (small-sample correction of the DM test)
- ID: **Identifier** (arXiv or DOI)
- LLM: **Large Language Model**
- LOTSA: **Large-scale Open Time Series Archive**
- MAE: **Mean Absolute Error**
- MASE: **Mean Absolute Scaled Error**
- MCS: **Model Confidence Set**
- MOMENT: **MOMENT**, a family of open time-series foundation models
- MSE: **Mean Squared Error**
- NVIDIA: **NVIDIA Corporation** (graphics processors)
- OOD: **Out Of Distribution**
- PID: **Proportional–Integral–Derivative** (controller)
- QLIKE: **Quasi-LIKelihood loss**
- RLHF: **Reinforcement Learning from Human Feedback**
- RV: **Realised Variance**
- TS: **Time Series** (in TS-Arena)
- TSA: **Time Series Analysis**
- VAT: **Value Added Tax**
- WQL: **Weighted Quantile Loss**

Today's question and route

- **Question:** can one pretrained model forecast series it has never seen, and how do we turn any forecast, pretrained or not, into intervals whose coverage we can guarantee or at least control?
 - ▶ two halves of one problem: point and quantile accuracy (foundation models), then honest uncertainty (conformal prediction)
- **Route** of the chapter
 - ▶ pretraining: data, tokens, patches, scaling laws; model families and design choices
 - ▶ benchmarks, contamination and testing across many series; evidence on Romanian and EU data; LLMs and their critique
 - ▶ conformal prediction: exchangeability, split conformal, CQR, the limits of conditional coverage
 - ▶ dependent data: weights, EnbPI, ACI, conformal PID; diagnostics; calibrating foundation-model intervals and VaR
- We build on [TSA](#), [Chapter 11](#) (Chronos, TimesFM, Moirai, zero-shot tests, CRPS and WQL), on [Chapter 1](#) (scoring rules, DM, MCS), [Chapter 9](#) (VaR backtests, ACI for VaR) and [Chapter 12](#) (deep architectures); Seminar 13 comes before this lecture

Learning outcomes

- Explain how a time series foundation model is pretrained: corpus, scaling, tokenisation or patching, output head and loss
- Compare model families and choose between zero-shot use, fine-tuning and in-context covariates
- Design a benchmark without leakage or contamination and test differences across many series with corrections for multiplicity
- Prove the coverage of split conformal and CQR, and state what conformal prediction cannot guarantee
- Apply weighted conformal, EnbPI, ACI and conformal PID to dependent data, diagnose coverage and calibrate foundation-model intervals

Reading, data and tools

- ▣ Foundation models: [Ansari et al. \(2024\)](#); [Ansari et al. \(2025\)](#); [Das et al. \(2024\)](#); [Woo et al. \(2024\)](#); [Auer et al. \(2025\)](#); benchmarks: [Aksu et al. \(2024\)](#); [Shchur et al. \(2025\)](#)
 - ▶ Conformal prediction: [Vovk, Gammernan and Shafer \(2022\)](#); [Angelopoulos and Bates \(2023\)](#); [Romano, Patterson and Candès \(2019\)](#); [Barber et al. \(2023\)](#); [Gibbs and Candès \(2021\)](#); [Angelopoulos, Candès and Tibshirani \(2023\)](#); [Xu and Xie \(2021\)](#)
 - ▶ Critique and evaluation: [Tan et al. \(2024\)](#); [Hewamalage, Ackermann and Bergmeir \(2023\)](#); survey of forecasting practice: [Petropoulos et al. \(2022\)](#)
- ▣ Python Quantlets of this chapter: [Quantlets/Ch_13](#)
 - ▶ open checkpoints on a CPU: [chronos-forecasting](#) (Chronos-Bolt, Chronos-2), [timesfm](#) (TimesFM 2.5), [tirex-ts](#) (TiRex); conformal methods, DM, MCS and backtests written in `numpy`
- ▣ Lecture notebook: [open in Google Colab](#)

Data used in this chapter

Series	Source	Sample
Romanian electricity load, hourly	Energy-Charts (ENTSO-E transparency data)	2022 – 30 September 2026
Bucharest air temperature, hourly	Open-Meteo historical archive	2022 – 2026
HICP annual inflation, 27 EU countries	Eurostat, monthly	2000 – August 2026
Realised variance, 5-minute returns	Bitcoin: Binance; S&P 500: Oxford-Man Heber et al. (2009)	2018 – 2026; 2000 – 2022
BET, S&P 500, NVIDIA, daily	EODHD daily closes	2000 – 2026

- Returns are 100 times log differences; the realised variance is in squared percent; every forecast uses only data available at its origin

Two lines of research that met



Royal Holloway, University of London: the Founder's Building

Photo: Diliff (2015); CC BY-SA 3.0; Wikimedia Commons

- 1990s–2005: conformal prediction at Royal Holloway: Vovk, Gammerman and Shafer, rooted in Kolmogorov's algorithmic randomness Vovk, Gammerman and Shafer (2005)
- 2002–2018: split (inductive) conformal Papadopoulos et al. (2002); distribution-free regression Lei et al. (2018); 2019: CQR Romano, Patterson and Candès (2019)
- 2021–2023: time series: ACI Gibbs and Candès (2021), EnbPI Xu and Xie (2021), conformal PID Angelopoulos, Candès and Tibshirani (2023), beyond exchangeability Barber et al. (2023)
- 2017–2025: Transformers Vaswani et al. (2017), foundation models Bommasani et al. (2021); for series: Lag-Llama, TimesFM, Moirai, Chronos (2023–2024), TiRex, Chronos-2 (2025)

Known from TSA and new here

- Known (TSA, Chapter 11): zero-shot use of Chronos, TimesFM, Moirai and Lag-Llama; mean scaling and quantisation in brief; pinball loss, CRPS, WQL, MASE; first warnings about contamination
- New: the research toolkit
 - ▶ design choices and their consequences (range limits, quantile heads, patching, covariates in context), scaling evidence
 - ▶ benchmarks with pre-registered windows after the model releases, tests across many series with Holm and BH corrections
 - ▶ conformal theory with proofs, its failure under dependence and the online methods that repair it, with coverage diagnostics
- Case studies: Ansari et al. (2024), Tan et al. (2024), Romano–Patterson–Candès (2019), Gibbs–Candès (2021), Barber et al. (2023), Angelopoulos–Candès–Tibshirani (2023), Xu–Xie (2021), with the designs of the papers, on our data

A foundation model as an amortised forecaster

- **Pretraining:** $\hat{\theta} = \arg \min_{\theta} \sum_{s \in \mathcal{D}_{\text{pre}}} \sum_t \ell(y_{t+1:t+H}^{(s)}, f_{\theta}(y_{t-C+1:t}^{(s)}))$ over a corpus $\mathcal{D}_{\text{pre}}$ of many series
 - ▶ C : context length; H : horizon; ℓ : a scoring rule (cross-entropy, pinball, negative log-likelihood)
 - ▶ the term **foundation model** [Bommasani et al. \(2021\)](#): trained once on broad data, then adapted to many tasks
- **Zero-shot** use: $\hat{\theta}$ fixed, the forecast for a new series is $f_{\hat{\theta}}(y_{T-C+1:T})$; no parameter is estimated on the target
 - ▶ the network performs the estimation step inside its forward pass: inference is amortised over the corpus
- A global model of [Chapter 12](#) is trained on the series of one data set; a foundation model on many data sets, frequencies and domains

Structure shared across series

- ▣ Shapes recur across domains: seasonal profiles, damped trends, level shifts, volatility clusters, intermittency
 - ▶ after scaling, a load curve and a web-traffic curve can look alike: the model learns a prior over shapes
- ▣ Not shared: the economics of a particular series (a tax change, a policy rule, a holiday calendar)
 - ▶ unless it is passed as a covariate in the context (Chronos-2) or learned by fine-tuning
- ▣ A no-free-lunch caveat: averaged over all processes no forecaster wins; pretraining helps only if the target resembles the corpus
 - ▶ financial returns are close to a martingale difference: little shape to transfer, much to overfit

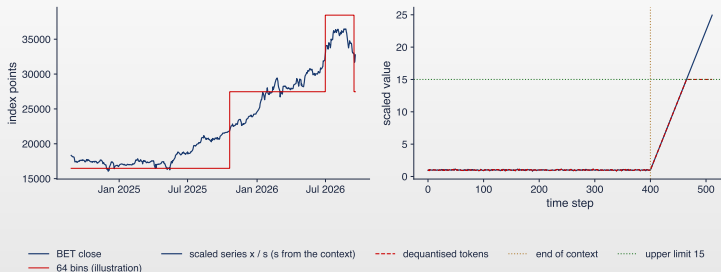
Pretraining data: scale and composition

- Chronos [Ansari et al. \(2024\)](#): public data sets plus two augmentations: **TSMixup** (convex mixtures of real series) and **KernelSynth** (series drawn from Gaussian processes with random composite kernels)
- TimesFM [Das et al. \(2024\)](#): about 10^{11} time points: Google Trends, Wikipedia page views, synthetic and public series; Moirai [Woo et al. \(2024\)](#): LOTSA, over 27 billion observations in nine domains
- Toto [Cohen et al. \(2025\)](#): observability metrics of a cloud provider, a corpus 4–10 times larger than those of earlier models; GIFT-Eval [Aksu et al. \(2024\)](#): a **non-leaking** pretraining set of about 230 billion points
- Composition matters more than size for us: economic and financial series are a small, low-frequency share of every corpus

Tokenisation: from real values to a vocabulary

- ▣ Chronos [Ansari et al. \(2024\)](#): **mean scaling** $\tilde{x}_t = x_t/s$, $s = \frac{1}{C} \sum_{t \leq C} |x_t|$ (context only), then **uniform bins** on $[-15, 15]$; 4096 tokens including special ones
 - ▶ the model is a language model (T5) trained with **cross-entropy**: the forecast is a categorical distribution over bins, sampled autoregressively
 - ▶ no notion of distance between tokens: neighbouring bins are as different as distant ones, unless the data teach otherwise
- ▣ Consequences: resolution 30s/4092; values above 15s cannot be predicted; scale-invariant by construction
- ▣ LLMTIME [Gruver et al. \(2023\)](#): digits as text tokens; Lag-Llama [Rasul et al. \(2023\)](#): lagged values as features; patch models: no vocabulary at all

The Chronos tokeniser on real data



- Left: BET closes, 27 August 2024 – 18 September 2026 ($C = 512$), and a 64-bin quantisation (illustration); right: a context near 1 followed by a rise to 25 times that level

Interpreting the tokeniser

- BET: $s = 23\,072$ points, bin width 169.1 points, maximum quantisation error 84.5 points: negligible for prices, but a fixed share of the level
- Right panel: the scale is set by the context, so the path that rises beyond 15s is clipped: 42% of the future steps are unreachable, the last one by 40%
- Explosive series (bubbles, [Chapter 16](#); hyperinflation) are outside the support of a mean-scaled vocabulary; Chronos-2 replaces it by an arcsinh transform and quantile outputs

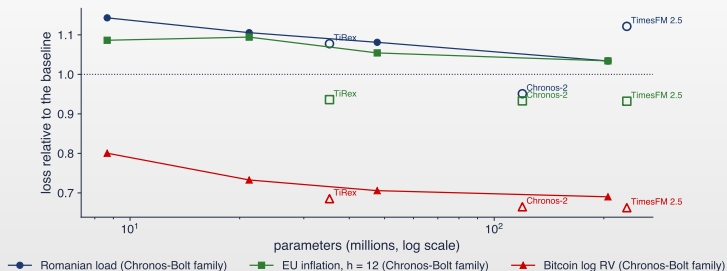
Patching and the output head

- ▣ **Patching** Nie et al. (2023): split the context into patches of P consecutive values, embed each patch as one token; attention costs $O((C/P)^2)$ instead of $O(C^2)$
 - ▶ TimesFM Das et al. (2024): input patches of 32, output patches of 128 (fewer autoregressive steps); Chronos-Bolt and Chronos-2: patches of 16; Moirai Woo et al. (2024): several patch sizes by frequency
- ▣ Output heads
 - ▶ categorical over bins (Chronos); **quantile head** trained by the pinball loss (Chronos-Bolt: 10%–90%; Chronos-2: 1%–99%; TimesFM 2.5, TiRex: deciles)
 - ▶ parametric: Student- t (Lag-Llama Rasul et al. (2023)), a mixture of distributions (Moirai)
- ▣ Direct multi-step quantiles (one pass for H steps) avoid error accumulation but give marginal, not joint, predictive distributions: path functionals (a maximum, a sum) need care

Scaling laws

- ▣ Language models [Kaplan et al. \(2020\)](#), [Hoffmann et al. \(2022\)](#): test loss falls as a power law, $L(N) \approx (N_c/N)^{\alpha_N}$, in the number of parameters N , data and compute, over orders of magnitude
- ▣ Time series: decoder-only Transformers show the same power laws in parameters, data and compute [Edwards et al. \(2024\)](#); the look-back length interacts with data size [Shi et al. \(2024\)](#)
 - ▶ out-of-distribution: the log-likelihood scales similarly in and out of distribution, but architecture matters; tweaks that help in distribution can reduce OOD scalability [Yao et al. \(2025\)](#)
- ▣ A scaling law is a statement about average loss on the corpus distribution, not about one Romanian series; small models (TiRex, 35 million parameters) beat larger ones on public leaderboards [Auer et al. \(2025\)](#)

Size and accuracy on our three tasks



- Losses relative to the task baseline: Romanian load (MAE / expert ARX, 2025–2026), EU inflation (MAE / random walk, $h = 12$, geometric mean over 27 countries), Bitcoin log RV (MSE / HAR); parameters counted in the loaded checkpoints

Interpreting size against accuracy

- ▣ Chronos-Bolt family (9–205 M parameters): load 1.143 $\rightarrow$ 1.034, inflation 1.086 $\rightarrow$ 1.034, Bitcoin 0.800 $\rightarrow$ 0.690
- ▣ Across families size is not the ranking: Chronos-2 (119 M) 0.951 on load; TiRex (35 M) 1.078; TimesFM 2.5 (231 M) 1.121
- ▣ Within the Chronos-Bolt family the loss falls with size on all three tasks (one exception: mini on inflation); across families architecture, corpus and training recipe change together, so three tasks are not a scaling study

Recap: pretraining

- ▣ Pretraining amortises estimation over a corpus; zero-shot use runs the estimator inside one forward pass
- ▣ Mean scaling and binning give scale invariance but a hard range; patching and quantile heads trade it for direct multi-step quantiles
- ▣ Scaling laws hold on average over corpora; on a given economic series, design and data matter more than size

The main open models

Model	Architecture	Parameters	Output
Chronos Ansari et al. (2024)	T5 encoder–decoder on tokens	20–710 M	categorical, sampled paths
Chronos-Bolt	T5 on patches of 16	9–205 M	9 quantiles, 10%–90%, 64 steps
Chronos-2 Ansari et al. (2025)	encoder with group attention	119 M	21 quantiles, 1%–99%; covariates
TimesFM 2.5 Das et al. (2024)	decoder-only, patches	231 M	mean and deciles
Moirai Woo et al. (2024)	masked encoder, any-variate attention	14–311 M	mixture distribution
Lag-Llama Rasul et al. (2023)	decoder-only on lag features	small	Student- t
MOMENT Goswami et al. (2024)	masked T5 encoder, multi-task	40–385 M	reconstruction; heads per task
TiRex Auer et al. (2025)	xLSTM	35 M	deciles

- Toto [Cohen et al. \(2025\)](#) (151 M, observability data) and Moirai 2.0 [Liu et al. \(2025\)](#) are open as well; TimeGPT [Garza et al. \(2023\)](#) is reached only through a paid API

The Chronos family

- ▣ **Chronos** Ansari et al. (2024): language-model recipe without changes; forecasts by sampling token paths; zero-shot results comparable to models trained on the target data (their Benchmark II)
- ▣ **Chronos-Bolt**: patch inputs, a direct quantile decoder; much faster; context up to 2048, 64 steps; quantiles only between 10% and 90%
 - ▶ a 95% interval or a VaR 1% cannot be read from its output: the request is clipped to the nearest trained level
- ▣ **Chronos-2** Ansari et al. (2025): group attention shares information across the series of a group (variates, related series, covariates); **in-context learning** of covariate effects
 - ▶ trained largely on synthetic multivariate structures imposed on univariate series; context 8192; 21 quantiles from 1% to 99%; arcsinh scaling

TimesFM, Moirai, Lag-Llama, MOMENT

- ▣ **TimesFM** [Das et al. \(2024\)](#): decoder-only, patches; output patch longer than input patch; trained on many granularities; version 2.5 adds a quantile head and long contexts
- ▣ **Moirai** [Woo et al. \(2024\)](#): masked encoder; frequency-specific patch sizes; any-variate attention flattens multivariate inputs; mixture output (Student- t , log-normal, negative binomial)
 - ▶ Moirai 2.0 [Liu et al. \(2025\)](#): a decoder-only simplification, smaller and better on GIFT-Eval
- ▣ **Lag-Llama** [Rasul et al. \(2023\)](#): a LLaMA-type decoder whose tokens are vectors of lagged values at many seasonal lags; Student- t head; strong after fine-tuning
- ▣ **MOMENT** [Goswami et al. \(2024\)](#): masked reconstruction on the Time series Pile; one encoder for forecasting, classification, anomaly detection and imputation

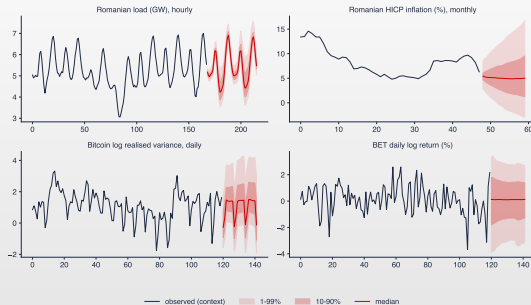
Recurrent again, observability, and a closed API

- ▣ **TiRex** Auer et al. (2025): an xLSTM (Chapter 12 recurrent cells with exponential gating and matrix memory) keeps a state across the context: state tracking for long horizons
 - ▶ contiguous patch masking (CPM) in training: the model learns to forecast with missing patches, i.e. several steps without feedback
- ▣ **Toto** Cohen et al. (2025): decoder-only for multivariate observability metrics; the BOOM benchmark (2807 series); shows that domain composition of the corpus drives results
- ▣ **TimeGPT** Garza et al. (2023): closed weights and undisclosed corpus behind a paid API
 - ▶ contamination cannot be audited; results are not reproducible if the service changes the model; data leave your institution

Zero-shot, fine-tuning, in-context covariates

- ▣ **Zero-shot:** $f_{\hat{\theta}}(y_{T-C+1:T})$; no training, no tuning risk; the only choices are C and the model
- ▣ **Fine-tuning:** θ initialised at $\hat{\theta}$, a few gradient steps on the target data (all weights, or adapters such as LoRA)
 - ▶ needs a validation split respecting time ([Chapter 12](#)); can forget the prior; with short economic series it rarely pays
- ▣ **In-context covariates** (Chronos-2): pass past values of x_t and future values of known regressors; the model infers their effect inside the forward pass
 - ▶ cross-learning: forecasting a batch of related series jointly (all 27 EU countries at one origin)
- ▣ Known future covariates must really be known at the origin: weather forecasts, not realised weather, unless an upper bound is the goal

Four zero-shot forecasts from one model



- Chronos-2 from the last origin of each data set: Romanian load (48 hours), Romanian HICP inflation (12 months), Bitcoin log realised variance and BET daily returns (22 days); bands 10–90% and 1–99%

Interpreting the four forecasts

- ▣ Load: the daily and weekly shapes are continued with narrow bands: shape transfer at its best
- ▣ Inflation: a smooth path towards the recent level with bands that widen with the horizon: close to what an AR model with persistence would give
- ▣ Log RV of Bitcoin: a weekly pattern (lower variance at weekends) around a level set by the context; returns: a flat median and a symmetric band, i.e. an unconditional distribution
- ▣ Return bands from the 1% and 99% quantiles: Chronos-2 VaR 1% for the next day is 3.90 (% of value); Section 8 checks whether such numbers are calibrated

Recap: model families

- ▣ Families differ in input (tokens, patches, lags), architecture (encoder, decoder, recurrent) and output (categorical, quantiles, parametric)
- ▣ Quantile range is a design constraint: deciles only, except Chronos-2
- ▣ Zero-shot is the default; covariates in context and cross-learning are the new levers; closed APIs cannot be audited

Benchmarks for pretrained models

- ▣ **GIFT-Eval** [Aksu et al. \(2024\)](#): 23 data sets, over 144 000 series, 177 million points, seven domains, ten frequencies, short to long horizons; a non-leaking pretraining set
- ▣ **fev-bench** [Shchur et al. \(2025\)](#): 100 tasks in seven domains, 46 with covariates; win rates and skill scores with bootstrap confidence intervals
- ▣ **Live benchmarks**: forecasts registered before the outcomes exist, e.g. TS-Arena [Meyer et al. \(2025\)](#) and Impermanent [Garza et al. \(2026\)](#): the only design immune to contamination by construction
- ▣ Classical references for what a fair comparison needs: M5 [Makridakis, Spiliotis and Assimakopoulos \(2022\)](#); pitfalls catalogued by [Hewamalage, Ackermann and Bergmeir \(2023\)](#)

Aggregating scores across series

- Scale-free scores: $\text{MASE} = \frac{1}{H} \sum_h |y_{T+h} - \hat{y}_{T+h}| / \frac{1}{T-m} \sum_t |y_t - y_{t-m}|$ Hyndman and Koehler (2006); WQL $= 2 \sum_{t,\tau} \rho_\tau(y_t - \hat{q}_{\tau,t}) / \sum_t |y_t|$
 - ▶ $\rho_\tau(u) = u(\tau - 1\{u < 0\})$ (pinball); the average over τ approximates the CRPS (Chapter 1)
- Relative scores $r_s = L_s(\text{model}) / L_s(\text{baseline})$ combined by the **geometric mean** Fleming and Wallace (1986): invariant to the choice of baseline in rankings, symmetric in gains and losses
 - ▶ skill score $1 - \bar{r}_{\text{geo}}$; win rate: share of series on which a model beats another
- The arithmetic mean of ratios rewards the baseline's weak series; the mean of raw errors is dominated by the series with the largest scale

Leakage and contamination

- ▣ **Series leakage:** an evaluation series (or a near copy) is in the pretraining corpus; Chronos separates in-domain (Benchmark I) from zero-shot (Benchmark II) results for this reason [Ansari et al. \(2024\)](#)
- ▣ **Temporal contamination:** the test window precedes the training cutoff; the model may have seen the outcomes of other, correlated series in the same period
 - ▶ LLMs recall exact economic values from before their cutoff [Lopez-Lira, Tang and Zhu \(2025\)](#); a two-data-set design against contamination for electricity prices [Pan and Ezzat \(2026\)](#)
- ▣ Designs that help
 - ▶ evaluate only after the release (or the declared cutoff) of every model in the comparison; we use 1 November 2025, after the last of our models (Chronos-2, 30 October 2025)
 - ▶ compare the relative skill before and after: a large drop after the release is a warning sign
 - ▶ pre-register origins, horizons, metrics and baselines before looking at the results

Testing across many series

- One series: DM with HLN correction [Diebold and Mariano \(1995\)](#), [Harvey, Leybourne and Newbold \(1997\)](#); conditional ability [Giacomini and White \(2006\)](#); many models: MCS [Hansen, Lunde and Nason \(2011\)](#) (all in [Chapter 1](#))
- S series tested separately: at 5% each, about $0.05S$ false rejections are expected under the null
 - ▶ **Holm** [Holm \(1979\)](#) (FWER): order $p_{(1)} \leq \dots \leq p_{(S)}$, reject while $p_{(k)} \leq \alpha / (S - k + 1)$
 - ▶ **Benjamini–Hochberg** [Benjamini and Hochberg \(1995\)](#) (FDR): reject the k^* smallest, $k^* = \max\{k : p_{(k)} \leq k\alpha/S\}$; valid under independence or positive dependence
- Pooled tests: average the loss differential across series at each origin, then one DM test with HAC variance: the cross-sectional correlation is kept in the time dimension
- Ranks over many data sets: Friedman and Nemenyi tests [Demšar \(2006\)](#); they ignore the size of differences and assume independent data sets

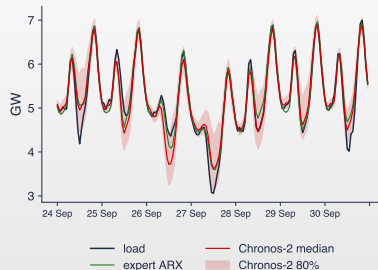
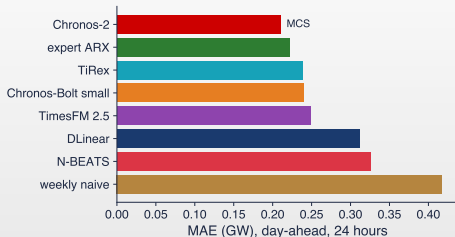
Recap: benchmarks and testing

- ▣ Aggregate scale-free relative scores by geometric means and report uncertainty over series
- ▣ Contamination is a property of the pair (model, test window): evaluate after the release or live
- ▣ Many series mean many tests: correct with Holm or BH, or pool before testing

The design of our comparisons

- ▣ **Romanian load:** day-ahead, 24 hours, every day from 1 January 2025 to 30 September 2026 (638 days); context 2048 hours for the foundation models
 - ▶ baselines: weekly naive; the expert ARX of Ziel and Weron (2018) for each hour (Chapter 1); DLinear Zeng et al. (2023) and N-BEATS Oreshkin et al. (2020) trained globally (Chapter 12)
- ▣ **EU inflation:** 27 countries, rolling monthly origins January 2012 – August 2025, $h = 1, \dots, 12$; random walk, $AR(p)$ by AIC, damped ETS Hyndman et al. (2002), DLinear and N-BEATS trained on the panel
- ▣ **Realised variance:** one day ahead, log RV; HAR Corsi (2009) on a rolling 1000-day window; QLIKE Patton (2011)
- ▣ **Foundation models:** Chronos-Bolt small, Chronos-2, TimesFM 2.5, TiRex, zero-shot, on a CPU; Moirai and Lag-Llama were not run (their libraries need older software versions)

Romanian load: day-ahead accuracy



- Left: MAE (GW) over 24 hours and 638 days, models in the 10% MCS marked; right: the last week of the sample, expert ARX and Chronos-2 with its 80% band

ATS_ch13_zero_shot

Interpreting the load comparison

- MAE: Chronos-2 0.210, expert ARX 0.221, TiRex 0.238, Chronos-Bolt 0.239, TimesFM 0.248; DLinear 0.311, N-BEATS 0.325, weekly naive 0.417 GW
- DM against the expert ARX: Chronos-2 $t = -3.05$ ($p = 0.002$); the other three foundation models are significantly worse; the 10% MCS is {Chronos-2}
- 80% interval coverage: Chronos-2 75.5%, Chronos-Bolt 81.2%, TimesFM 81.2%, expert ARX 78.1%: none is far off, none is exact
- A univariate zero-shot model matches an established domain model on a series with strong, regular shapes; the global deep models trained on three years lose to both

In-context covariates: calendar and temperature

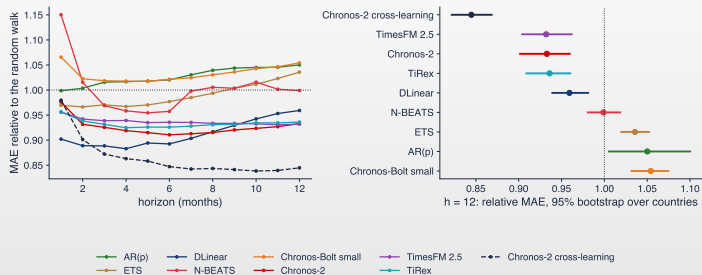


- Chronos-2 MAE on all days, on ordinary days and on the 29 public holidays: univariate, with weekend and holiday flags, and with flags plus Bucharest hourly temperature (realised values for the forecast day: an upper bound)

Interpreting the covariate experiment

- ▣ Holidays: MAE 0.527 univariate, 0.396 with the calendar, 0.365 with temperature; ordinary days: 0.195, 0.193, 0.178
- ▣ Calendar against univariate: DM $t = -2.79$ ($p = 0.005$); temperature on top: $t = -7.02$ ($p < 0.001$)
- ▣ The holiday effect is learned in context from a few earlier holidays in the 2048-hour window; no parameter was estimated
- ▣ Realised temperature is not available at the origin: an operational test needs archived weather forecasts

EU inflation: 27 countries, 12 horizons

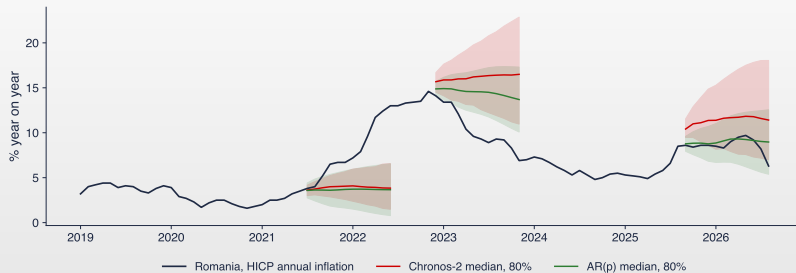


- Left: MAE relative to the random walk by horizon, geometric mean over countries; right: $h = 12$ with 95% bootstrap intervals over countries; origins January 2012 – August 2025

Interpreting the inflation panel

- ▣ $h = 1$: AR 0.999, Chronos-2 0.976, TimesFM 0.956, TiRex 0.956; $h = 12$: AR 1.050, ETS 1.036, Chronos-2 0.933, TimesFM 0.932, TiRex 0.936, Bolt 1.054
- ▣ Global deep models at $h = 12$: DLinear 0.959, N-BEATS 0.999; Chronos-2 with cross-learning across the 27 countries: 0.845
- ▣ CRPS relative to AR (geometric mean over countries): Chronos-2 0.860, TimesFM 0.875, TiRex 0.865, ETS 0.972
- ▣ The 2021–2023 surge dominates every average: no univariate method foresaw it, so differences are about the speed of adaptation, not about foresight

Romania: forecasts from three origins

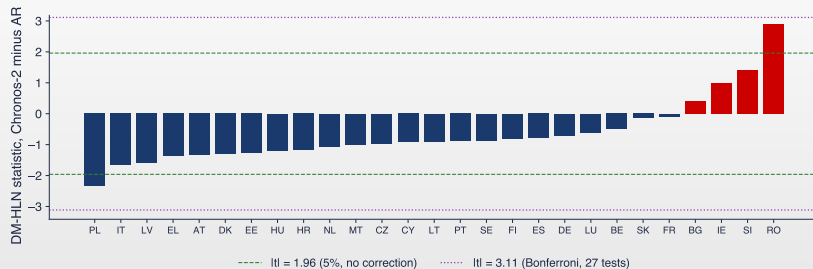


- Romanian HICP annual inflation and 12-month forecasts with 80% bands from Chronos-2 and AR(p), from June 2021, November 2022 and August 2025

Interpreting the Romanian paths

- From June 2021: 12 months later inflation was 13.0%; Chronos-2 median 3.8% (80%: 1.5 – 6.6), AR 3.7%
- From November 2022, near the peak (14.6% in November 2022): outcome 6.9%; Chronos-2 16.5%, AR 13.7%
- From August 2025: outcome 6.3%; Chronos-2 11.4%, AR 9.0%; the latest observation is 6.3% (August 2026)
- Both methods extrapolate persistence; turning points come from information outside the series (energy prices, the end of the price caps, the VAT increase of August 2025, whose effect on annual inflation lasts exactly 12 months): known in advance, hence natural covariates

Twenty-seven tests at once

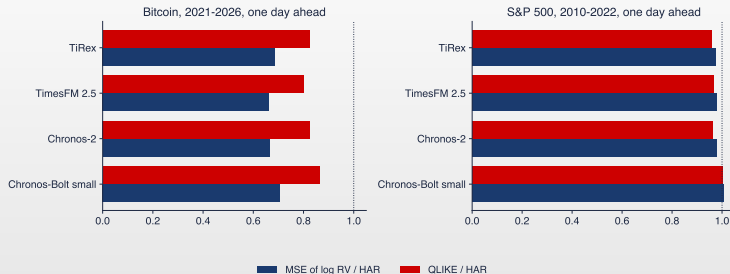


- Country-level DM-HLN statistics (HAC with $h - 1$ lags) of the absolute errors of Chronos-2 minus $AR(p)$ at $h = 12$; negative: Chronos-2 better

Interpreting the multiple tests

- 23 of 27 statistics are negative; 2 are significant at 5% without correction (1 in favour of Chronos-2, 1 in favour of AR)
- With Holm: 0 rejections; with Benjamini–Hochberg: 0; Romania: $t = 2.88$, $p = 0.005$
- Pooled over the 27 countries (average loss differential at each origin, HAC variance): $t = -1.35$, $p = 0.180$; the bootstrap intervals of the previous chart treat countries as independent and look sharper than they are
- A paper that reports the countries where its model wins at 5% reports noise; the pooled test and the multiplicity-adjusted counts are the honest summary

Realised variance: foundation models against HAR

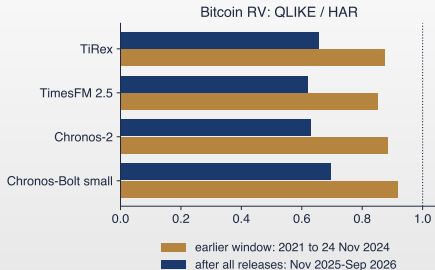
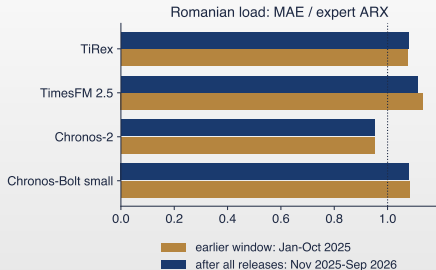


- One-day-ahead log RV: MSE and QLIKE relative to HAR; Bitcoin (Binance, 1 January 2021 – 18 September 2026, $n = 2084$), S&P 500 (Oxford-Man, 4 January 2010 – 25 February 2022, $n = 3047$)

Interpreting the volatility comparison

- Bitcoin, QLIKE relative to HAR: Chronos-2 0.825, TimesFM 0.800, TiRex 0.824, Bolt 0.864; MCS (10%): {Chronos-2, TimesFM 2.5, TiRex}
- S&P 500: Chronos-2 0.963, TimesFM 0.968, TiRex 0.956, Bolt 1.004; MCS (10%): {Chronos-2, TimesFM 2.5, TiRex}
- Three of the four zero-shot models beat HAR on both assets and HAR is outside both MCS: the slowly decaying memory of log RV ([Chapter 10](#)) is a shape a corpus can teach; the gain is large for Bitcoin, small for the S&P 500
- Recent studies reach mixed verdicts [Goel et al. \(2025\)](#), [Brini \(2026\)](#), [Rahimikia, Ni and Wang \(2025\)](#): gains depend on the asset, the horizon and the loss, and fine-tuning or pretraining on financial data often matters

Before and after the releases



- Relative loss of each model in an earlier window and in the window after every release (from 1 November 2025): load (MAE / expert ARX) and Bitcoin (QLIKE / HAR)

Interpreting the contamination check

- Load, Chronos-2: 0.952 before, 0.950 after; TimesFM: 1.133, 1.112; the earlier window (January–October 2025) precedes both releases
- Bitcoin, Chronos-2: 0.886 (2021 – 24 November 2024) and 0.631 (after); TiRex: 0.874, 0.658; post-release windows have 334 and 322 days
- No deterioration after the releases (for Bitcoin the ratios are even lower): no evidence of contamination here; but the absence of a drop does not prove absence of leakage, and markets change between windows

Recap: the evidence

- ▣ Strong shapes (load): the best foundation model matches or beats the domain model; weak shapes (returns): little to transfer
- ▣ Inflation: ahead of AR and ETS in the panel average (most with cross-learning), but no country difference survives Holm or BH and the pooled test is not significant
- ▣ Realised variance: zero-shot models beat HAR, clearly for Bitcoin, narrowly for the S&P 500
- ▣ Covariates in context help where the calendar matters; post-release windows show no sign of contamination

LLMTime: numbers as text

- ▣ Gruver et al. (2023): rescale, round to a fixed number of digits, write values as digits separated by spaces and commas; an off-the-shelf LLM continues the string
 - ▶ digit-by-digit probabilities define a hierarchical (multi-bin) density over continuous values; repeated sampling gives quantiles
- ▣ Findings of the paper: competitive zero-shot on some benchmarks; tokenisation of numbers matters (a newer model could do worse because of how it splits digits); alignment (RLHF) hurts calibration
- ▣ Costs: thousands of tokens per series and many samples per forecast; the context window limits the history; the LLM may have memorised the data

Reprogramming a frozen LLM

- ▣ **Time-LLM** [Jin et al. \(2024\)](#): patches of the series are mapped onto text prototypes (word embeddings), a prompt prefix describes the task and statistics; the LLM stays frozen, a projection reads the forecast
- ▣ **One Fits All** (GPT4TS) [Zhou et al. \(2023\)](#): a frozen GPT-2 with trained input embedding, layer norms and output layer, for forecasting, classification and anomaly detection
- ▣ The claim: knowledge from language transfers to series; the test of the claim is an ablation that removes the language model

Case study: are language models actually useful?

- Tan et al. (2024): three LLM-based forecasters (including Time-LLM and GPT4TS), the same benchmarks, three ablations
 - remove the LLM; replace it by one attention layer; replace it by a basic Transformer block
- Ablations
 - remove the LLM; replace it by one attention layer; replace it by a basic Transformer block
- Result: no degradation, often an improvement; pretrained LLMs do no better than the same model trained from scratch, do not model sequential dependence, do not help in few-shot settings
- Training and inference cost orders of magnitude more
- The method matters more than the result
- an architecture claim needs an ablation with matched compute and tuning
- time-series foundation models trained on series (Chronos, TimesFM, TiRex) are not affected by this critique; LLMs used as forecasters are
- LLMs remain useful around forecasting: code, literature, critique (Section 9)

Memorisation and look-ahead

- ▣ [Lopez-Lira, Tang and Zhu \(2025\)](#): LLMs reproduce macroeconomic and market values from their training period with high precision, even when asked not to use future information
 - ▶ a backtest before the training cutoff measures recall, not forecasting
- ▣ For time-series foundation models the same logic applies to public series in the corpus (electricity, traffic, M4)
 - ▶ our evidence: windows after every release (Section 4); the strongest evidence: live, pre-registered forecasts
- ▣ A practical rule: state the training cutoff of every model and the start of the evaluation window in the same table

Recap: LLMs

- ▣ LLMTime and reprogramming show that text models can be bent to series; ablations show the language part is rarely what helps
- ▣ Memorisation makes pre-cutoff evaluation of LLM forecasts invalid

Where conformal prediction comes from



Alexander Gammerman, Royal Holloway, 2018

Photo: Jaguar224 (2018); CC BY-SA 4.0; Wikimedia Commons

- Vovk, Gammerman and Shafer Vovk, Gammerman and Shafer (2005), Vovk, Gammerman and Shafer (2022): prediction sets valid in finite samples under exchangeability only, for any predictive model
- The idea: how "conforming" a candidate value is to the data seen, measured by a rank
- Modern statistics adopted it as a wrapper around machine learning Lei et al. (2018), Angelopoulos and Bates (2023)

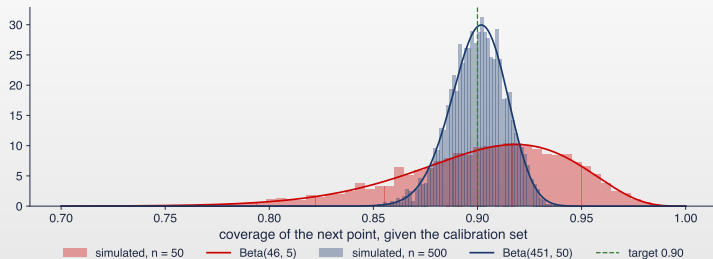
Exchangeability and the target guarantee

- $Z_1, \dots, Z_{n+1}$, $Z_i = (X_i, Y_i)$, are **exchangeable** if $(Z_{\pi(1)}, \dots, Z_{\pi(n+1)}) \stackrel{d}{=} (Z_1, \dots, Z_{n+1})$ for every permutation π
 - ▶ i.i.d. implies exchangeable; draws without replacement are exchangeable but dependent; a stationary AR(1) is not exchangeable
- **Marginal coverage**: $\Pr\{Y_{n+1} \in \hat{C}(X_{n+1})\} \geq 1 - \alpha$, the probability taken over the calibration data and the test point together
 - ▶ not **conditional** coverage $\Pr\{Y \in \hat{C}(x) \mid X = x\} \geq 1 - \alpha$ for every x , and not coverage given the calibration sample
- A **nonconformity score** $s(x, y)$: large when y is unusual given x , e.g. $|y - \hat{\mu}(x)|$, $|y - \hat{\mu}(x)|/\hat{\sigma}(x)$, or the CQR score below

Split conformal prediction

- Fit $\hat{\mu}$ on a training set; compute $S_i = s(X_i, Y_i)$ on n calibration points [Papadopoulos et al. \(2002\)](#), [Lei et al. \(2018\)](#)
 - ▶ $\hat{q} = S_{(k)}$, $k = \lceil (n+1)(1-\alpha) \rceil$ (the k -th smallest; $+\infty$ if $k > n$); $\hat{C}(x) = \{y : s(x, y) \leq \hat{q}\}$
- **Theorem:** if $(X_i, Y_i)_{i \leq n+1}$ are exchangeable, $\Pr\{Y_{n+1} \in \hat{C}(X_{n+1})\} \geq 1 - \alpha$; with no ties, also $\leq 1 - \alpha + 1/(n+1)$
 - ▶ proof: given $\hat{\mu}$, the scores $S_1, \dots, S_{n+1}$ are exchangeable, so the rank of S_{n+1} is uniform on $\{1, \dots, n+1\}$; $Y_{n+1} \in \hat{C} \iff \text{rank} \leq k$, probability $k/(n+1) \geq 1 - \alpha$ (▶ [Appendix](#))
- No assumption on the model or on the distribution: a bad model gives wide, not invalid, intervals

Coverage given the calibration set



- Coverage $F(\hat{q})$ of the next point for 4000 calibration sets of $n = 50$ and $n = 500$ scores, $\alpha = 0.1$, and the exact law Beta(k , $n + 1 - k$)

Interpreting the coverage distribution

- Mean coverage 0.902 ($n = 50$) and 0.900 ($n = 500$), as the theory gives (0.902, 0.900): the marginal guarantee holds on average over calibration sets
- One calibration set is one draw: s.d. 0.041 and 0.014; with $n = 50$, 26% of the sets give coverage below 0.88, with $n = 500$ only 7%
- Since $F(\hat{q}) \sim \text{Beta}(k, n + 1 - k)$, choosing n is a sample-size calculation: the s.d. is about $\sqrt{\alpha(1 - \alpha)/n}$

Conformalized quantile regression

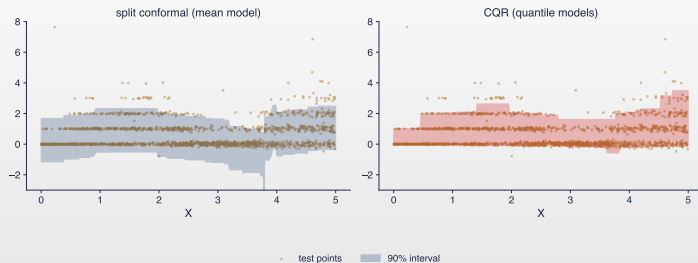


Emmanuel Candès, 2012

Photo: Renate Schmid, MFO (2012); CC
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- Romano, Patterson and Candès (2019): fit quantile regressions Koenker and Bassett (1978) $\hat{q}_{\alpha/2}(x)$, $\hat{q}_{1-\alpha/2}(x)$ on the training set
- Score $s(x, y) = \max\{\hat{q}_{\alpha/2}(x) - y, y - \hat{q}_{1-\alpha/2}(x)\}$: negative inside the band
- $\hat{C}(x) = [\hat{q}_{\alpha/2}(x) - \hat{q}, \hat{q}_{1-\alpha/2}(x) + \hat{q}]$: the band is shifted outwards (or inwards if $\hat{q} < 0$)
- Marginal coverage by the split theorem; the width adapts to heteroskedasticity through the quantile models
- The same score calibrates the quantiles of any foundation model (Section 8)

CQR on the design of Romano, Patterson and Candès



- $Y = \text{Pois}(\sin^2 X + 0.1) + 0.03X\epsilon_1 + 25 \cdot 1\{U < 0.01\}\epsilon_2$, $X \sim U[0, 5]$ (their Figure 1); 1000 training and 1000 calibration points; gradient boosting for the mean and for the 5% and 95% quantiles

Interpreting CQR

- Test coverage: split conformal 0.918, CQR 0.915, the raw quantile models 0.883; mean width 2.84, 2.20 and 2.17
- The raw quantile regressions undercover; CQR repairs them at almost no extra width, while the constant-width split band is 29% wider
- By X bins both stay within a few points of 90% (worst bins: CQR 0.892, split 0.896); the gain of CQR is in width: narrow where Y is concentrated, wide where the Poisson noise is large

The limits of conditional coverage

- **Impossibility** Vovk (2012), Lei and Wasserman (2014), Foygel Barber et al. (2021): if $\hat{C}$ has conditional coverage $\geq 1 - \alpha$ at almost every x for every distribution with a continuous X , then its expected length is infinite
 - ▶ no finite-sample method can certify coverage at each point of a continuous covariate without assumptions
- What is achievable: coverage within a finite set of groups (Mondrian conformal: calibrate separately by group); approximate conditional coverage under smoothness; coverage conditional on the calibration set with high probability (Beta law)
- For time series the relevant condition is the past: $\Pr\{Y_t \in \hat{C}_t \mid \mathcal{F}_{t-1}\}$; a VaR backtest (Chapter 9) tests exactly this property for one-sided sets

Recap: conformal prediction

- Split conformal: a rank argument gives finite-sample marginal coverage under exchangeability, for any model
- CQR adds adaptivity; the calibration sample size controls the variability of realised coverage
- Conditional coverage is impossible in general: test it, do not claim it

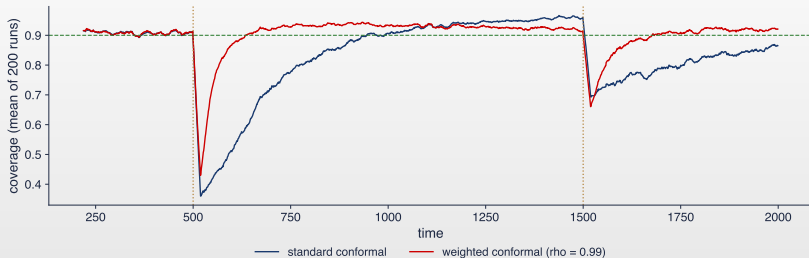
Exchangeability violated: what survives

- ▣ Time series violate exchangeability three ways: serial dependence, changing volatility, structural change; the last two break coverage most
- ▣ **Stationary and mixing:** split conformal remains approximately valid, with a coverage gap controlled by β -mixing coefficients and the calibration size [Oliveira et al. \(2024\)](#)
 - ▶ block permutations restore exactness under weaker conditions [Chernozhukov, Wüthrich and Zhu \(2018\)](#)
- ▣ **Non-stationary:** no static method can work; the threshold must move with the data: weights (Barber et al.), refitting (EnbPI), feedback on errors (ACI, PID)
- ▣ A common online frame: scores $S_t = s(X_t, Y_t)$ computed from out-of-sample forecasts; at t choose a threshold q_t from $S_1, \dots, S_{t-1}$; miss $\text{err}_t = 1\{S_t > q_t\}$

Conformal prediction beyond exchangeability

- Barber et al. (2023): fixed weights $w_i \in [0, 1]$, chosen before seeing the data; $\hat{q}$ = the $(1 - \alpha)$ quantile of $\sum_i \tilde{w}_i \delta_{S_i} + \tilde{w}_{n+1} \delta_{+\infty}$, $\tilde{w}_i = w_i / (1 + \sum_j w_j)$, $w_{n+1} = 1$
 - **Theorem:** coverage $\geq 1 - \alpha - \sum_i \tilde{w}_i d_{TV}(Z, Z^i)$, where Z^i swaps the test point with point i
- Under exchangeability the gap is zero; under drift, d_{TV} is small for recent points, so geometric weights $w_i = \rho^{n+1-i}$ keep it small
 - effective sample size $\approx 1/(1 - \rho)$: $\rho = 0.99$ uses about 100 recent scores; robustness against variance of the threshold
- Covariate shift with known likelihood ratio: weights $w(x) = dP_{\text{test}}/dP_{\text{train}}$ give exact coverage Tibshirani et al. (2019)

Weighted conformal under changepoints



- $Y_t = X_t' \beta_t + \varepsilon_t$, $X_t \sim N(0, I_d)$, β changes at $t = 500$ and $t = 1500$; least squares on all past data; prequential absolute residuals as scores; coverage averaged over 200 runs (20-step moving average)

Interpreting the weighted method

- Average coverage after the burn-in: standard 0.836, weighted ($\rho = 0.99$) 0.900; worst 20-step average 0.36 against 0.43
- Both collapse at a break, because the least-squares fit is wrong for a while; the weighted threshold forgets the old scores and recovers within about 100 steps
- Price: mean width 7.47 against 6.22; the guarantee is a bound in terms of d_{TV} , not exact coverage

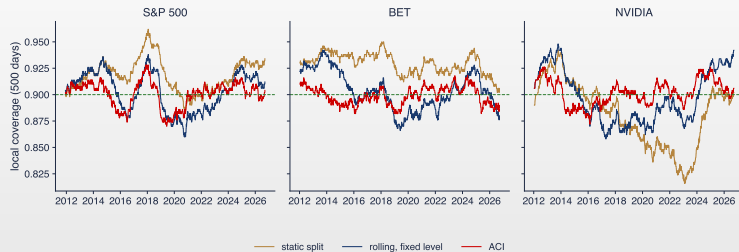
EnbPI: ensembles without data splitting

- ▣ Xu and Xie (2021), Xu and Xie (2023): fit B models on bootstrap samples (blocks, to respect dependence) of the training period
 - ▶ leave-one-out residuals: for point i , aggregate only the models whose bootstrap sample excluded i
 - ▶ interval at t : $\hat{f}(x_t) \pm$ the $(1 - \alpha)$ quantile of the last T absolute residuals; after y_t is observed, its residual enters the window and the oldest leaves
- ▣ No refitting at each step, no calibration split: efficient for long streams
- ▣ Guarantee: asymptotic, conditional coverage if the errors are stationary and strongly mixing and the ensemble is consistent; designed for energy series (solar and wind)

Adaptive conformal inference

- **Gibbs and Candès (2021)**: use level α_t at time t , $q_t =$ the conformal $(1 - \alpha_t)$ quantile of recent scores, then $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t)$
 - ▶ a miss lowers α_t (wider sets); a hit raises it; $\alpha_t \leq 0$ gives $q_t = +\infty$ (the whole line), $\alpha_t \geq 1$ the empty set
- **Theorem**: for any sequence of data, $\left| \frac{1}{T} \sum_{t=1}^T \text{err}_t - \alpha \right| \leq \frac{\max\{\alpha_1, 1 - \alpha_1\} + \gamma}{\gamma T}$
 - ▶ proof: α_t stays in $[-\gamma, 1 + \gamma]$ and $\alpha_{T+1} - \alpha_1 = \gamma \sum_t (\alpha - \text{err}_t)$; divide by γT
- Long-run frequency, not conditional coverage: an adversary-proof average. Choosing γ adaptively: AgACI Zaffran et al. (2022), DtACI Gibbs and Candès (2024); ACI for VaR: Chapter 9

ACI on stock-market volatility

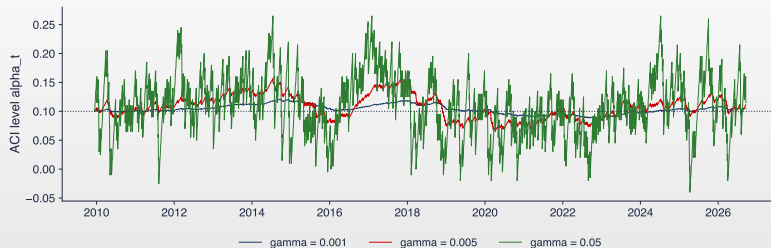


- Design of Gibbs and Candès (2021): $V_t = r_t^2$, $\hat{\sigma}_t^2$ from a GARCH(1,1) fitted on the previous 1250 days, score $|V_t - \hat{\sigma}_t^2|/\hat{\sigma}_t^2$, $\alpha = 0.1$, $\gamma = 0.005$; local coverage over 500 days; evaluation from 15 December 2009

Interpreting ACI on volatility

- ▣ Overall coverage, S&P 500: static 0.918, rolling 0.903, ACI 0.901; BET: 0.928, 0.903, 0.899; NVIDIA: 0.882, 0.900, 0.901
- ▣ Local coverage of the static method ranges from 0.884 to 0.962 (S&P 500); ACI stays within 0.870 – 0.928
- ▣ Christoffersen conditional coverage of the misses (S&P 500): static $p < 0.001$, ACI $p = 0.968$: a calibration fixed on 2005–2009 is too wide afterwards; updating restores both the frequency and the independence of misses

The step size of ACI



- S&P 500 scores of the previous chart: the level α_t for $\gamma = 0.001, 0.005$ and 0.05

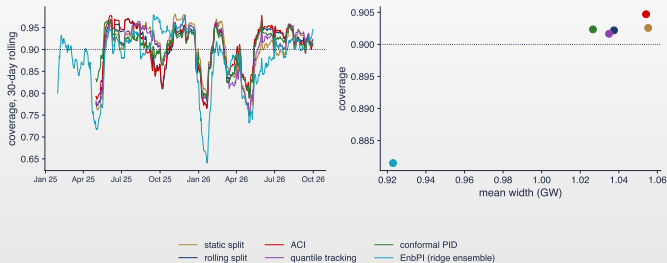
Interpreting the step size

- Miss rates 0.0987, 0.0994, 0.0999 against the bounds 0.2138, 0.0430, 0.0045 ($T = 4214$): the theorem holds with room to spare
- Large γ : α_t ranges $-0.04 - 0.27$, infinite intervals on 1.6% of days; small γ : slow reaction, long runs of misses after a shock
- γ trades adaptivity against stability: tune it on a past window, or let DtACI aggregate several values

Conformal PID control

- Angelopoulos, Candès and Tibshirani (2023): **quantile tracking** (P): $q_{t+1} = q_t + \eta(\text{err}_t - \alpha)$, i.e. online gradient descent on the pinball loss of the score
 - **Proposition**: if $S_t \in [0, B]$, then $|\frac{1}{T} \sum_t (\text{err}_t - \alpha)| \leq (B + \eta)/(\eta T)$: the threshold moves on the scale of the scores, not of α
- **Integrator** (I): add $r_t(\sum_{i \leq t} (\text{err}_i - \alpha))$, $r_t(x) = K_I \tan(x \log t / (t C_{\text{sat}}))$: reacts to accumulated coverage error, saturates to keep the guarantee
- **Scorecaster** (D-like): a model that forecasts the next score from its past (seasonality, trends in the errors) and is added to the threshold
- ACI is the special case that tracks the level instead of the quantile; PID borrows its vocabulary from control engineering: proportional, integral, derivative

Online methods on Romanian load



- 90% day-ahead intervals around the Chronos-Bolt median, one score stream per hour (24 streams), calibration window 91 days; EnbPI on the expert ARX regressors (ridge, 20 block-bootstrap models); evaluation from 2 April 2025

Q.ATS_ch13_conformal_time

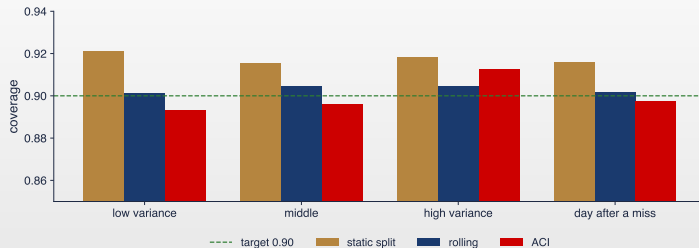
Interpreting the online methods

- ▣ Coverage: static 0.903, rolling 0.902, ACI 0.905, quantile tracking 0.902, PID 0.902, EnbPI 0.881
- ▣ Mean width (GW): 1.06, 1.04, 1.05, 1.03, 1.03, 0.92; worst 30-day coverage: static 0.747, PID 0.787
- ▣ Over the whole period every method except EnbPI is close to 90%; the differences are in the dips after seasonal transitions and in width, where PID is best on both counts
- ▣ EnbPI centres the band on its own ridge ensemble, less accurate than the base forecast: narrower intervals, lower coverage

Coverage diagnostics

- **Unconditional:** miss rate and Kupiec [Kupiec \(1995\)](#); **independence** of misses: Christoffersen [Christoffersen \(1998\)](#) ([Chapter 9](#))
 - ▶ regression check in the spirit of the DQ test [Engle and Manganelli \(2004\)](#): regress $\text{err}_t - \alpha$ on lagged misses and on the predicted width
- **Local:** rolling coverage; coverage by bins of a variable known at the origin (predicted volatility, hour, regime): the empirical counterpart of conditional coverage
- **Sharpness:** mean width and the interval score $W + \frac{2}{\alpha}(\ell - y)1\{y < \ell\} + \frac{2}{\alpha}(y - u)1\{y > u\}$ [Winkler \(1972\)](#), [Gneiting and Raftery \(2007\)](#), a proper score for central intervals
- Report all three: a method that is valid but twice as wide is not a better method

Coverage by volatility regime



- S&P 500, 90% volatility intervals of the ACI chart: coverage by tercile of the GARCH variance forecast and on the day after a miss

Interpreting conditional coverage

- ▣ Static: 0.921 (low variance), 0.918 (high), 0.916 after a miss; ACI: 0.893, 0.912, 0.897
- ▣ Coverage is almost flat across volatility terciles and after a miss, for every method: dividing by the GARCH variance makes the score close to pivotal
- ▣ Contrast: with the raw score $|r_t|$ (Seminar 13, B3) misses cluster in turbulent years whatever the calibration; conditional coverage is a property of the score, not of the conformal step

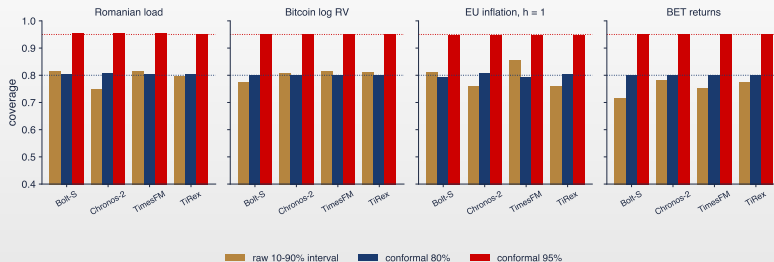
Recap: conformal methods for time series

- ▣ Weighted conformal: a coverage bound in total variation; geometric weights adapt to drift
- ▣ EnbPI: bootstrap ensembles and a sliding residual window, asymptotic validity under mixing
- ▣ ACI and PID: feedback on misses gives long-run coverage for any sequence; conditional coverage must still be tested

The need to calibrate foundation-model quantiles

- ▣ Pretrained quantiles are calibrated on the corpus distribution, not on your series: miscalibration is the default, in either direction
- ▣ Most models output only 10%–90%: a 95% interval or VaR 1% requires extrapolation beyond the trained levels
- ▣ Recipe: online CQR on the model's own band, score $S_t = \max\{\hat{q}_{0.1,t} - y_t, y_t - \hat{q}_{0.9,t}\}$; ACI chooses the threshold for any target level (80% or 95%)
 - ▶ one-sided version for VaR: $S_t = \hat{q}_{0.1,t} - y_t$, threshold at level α ; the VaR is $-(\hat{q}_{0.1,t} - q_t)$
- ▣ Further reading on conformal VaR recalibration of foundation models: [Pele et al. \(Conformal_Oracle\)](#); [Pele et al. \(TSFM_VaR_CEE\)](#)

Raw and conformal coverage of foundation models

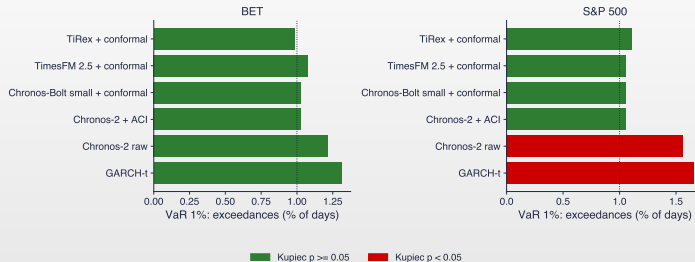


- Coverage of the raw 10–90% band and of the online-CQR 80% and 95% intervals (ACI, window 250, $\gamma = 0.005$): Romanian load (24 hourly streams), Bitcoin log RV, EU inflation at $h = 1$ (27 streams), BET daily returns

Interpreting the calibration

- ▣ Raw 80% bands: load 75–81%, Bitcoin 77–81%, inflation 76–85%, BET returns 72–78% across the four models
- ▣ After online CQR: 80% intervals cover 79–81%, 95% intervals 94.5–95.4% in every task
- ▣ Raw bands miss in both directions (too narrow: Chronos-2 on load; too wide: TimesFM on inflation); the conformal step widens or narrows them, with a mean width between 0.90 and 1.24 times the raw width, and reaches 95% from models that output only deciles

Value at Risk from foundation models



- VaR 1% exceedances, BET and S&P 500 from 28 December 2017 (2175 days): GARCH- t (rolling 1000 days, [Chapter 9](#)), Chronos-2 raw and with ACI, deciles of the other models extended by a one-sided conformal shift

Interpreting the VaR backtest

- ▣ BET, VaR 1%: GARCH- t 1.31% (Kupiec $p = 0.170$), Chronos-2 raw 1.22% ($p = 0.331$), Chronos-2 + ACI 1.03% ($p = 0.893$), TimesFM + conformal 1.08%
- ▣ S&P 500: GARCH- t 1.66% (Kupiec $p = 0.007$), Chronos-2 raw 1.56% ($p = 0.021$), + ACI 1.06% ($p = 0.802$, Christoffersen $p = 0.454$)
- ▣ Calibration fixes the exceedance rate of every model; it does not add tail information: the conformally extended deciles have a larger quantile loss than GARCH- t (BET, Chronos-Bolt: DM $t = 2.70$), while Chronos-2, trained on 1% quantiles, does not ($t = 0.10$)

Recap: calibration

- ▣ Treat foundation-model quantiles as scores to be calibrated, not as probabilities
- ▣ Online CQR with ACI gives target coverage at any level, including beyond the trained quantiles
- ▣ Backtest calibrated VaR as in [Chapter 9](#): frequency, independence and loss

An open question

- Do time-series foundation models beat domain-specific econometric models on Central and Eastern European data after their release dates, once multiplicity is accounted for?
 - ▶ formal: H_0 : equal expected loss (pooled over pre-registered series) of the best foundation model and the best domain model, in a window that starts after every model release
 - ▶ falsified by a pooled DM test at 5% and an MCS that excludes one side, on series and horizons fixed before the data arrive
- Why it matters: central banks and grid operators consider replacing tuned models by zero-shot ones; published wins often come from a favourable choice of series
 - ▶ literature to start from: [Aksu et al. \(2024\)](#), [Shchur et al. \(2025\)](#), [Meyer et al. \(2025\)](#), [Pan and Ezzat \(2026\)](#), [Brini \(2026\)](#)

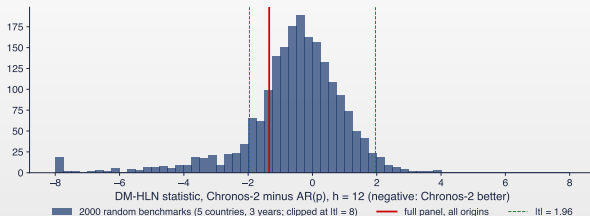
The discovery loop with an AI assistant

- An AI assistant (an LLM such as Claude, ChatGPT, Gemini or Copilot) speeds up each step; Semantic Scholar and Elicit help with the literature
 - ▶ **literature:** List peer-reviewed or arXiv papers since 2024 that evaluate time-series foundation models only after their release dates; give arXiv IDs or DOIs. Then check every identifier
 - ▶ **hypothesis:** Write a pre-registration: series, origins, horizons, losses, baselines, the pooled test and the MCS level.
 - ▶ **code and replication:** ask for a wrapper that returns quantiles at fixed levels for each model; replicate a known number first (the Chronos-Bolt quantile range, our load MAE)
 - ▶ **critique:** Act as a hostile referee: list every way this benchmark could favour the foundation models.
- Report: what was asked, what was kept, what was rejected (AI_USE.md, AI_ERRORS.md)

What the human checks

- ▣ Every reference exists and says what is claimed (arXiv ID or DOI resolves, the title matches, the result is in the paper)
- ▣ Release dates and training cutoffs of every model, from the model cards, against the start of the evaluation window
- ▣ Quantile levels actually produced by each model (a request for 1% may silently return 10%)
- ▣ Baselines tuned with the same care; the context available to every model is the same
- ▣ Multiplicity: number of series, horizons and models tested; the pooled test and adjusted p -values reported

Mini-case: how much can the choice of benchmark change the verdict?



- DM-HLN statistics of Chronos-2 against $AR(p)$ at $h = 12$ on 2000 random benchmarks of 5 EU countries and 3 years of origins, and on the full panel
- 11.2% of the random benchmarks declare Chronos-2 significantly better and 2.1% significantly worse; the full panel gives $t = -1.35$ ($p = 0.180$). An AI summary of one such paper as evidence for (or against) foundation models is wrong; only the pre-registered panel answers

Project idea

- ▣ **A pre-registered, post-release benchmark of foundation models for CEE macro and risk, with conformal calibration**
 - ▶ replicate: the zero-shot protocol of [Ansari et al. \(2024\)](#) (WQL, MASE, geometric means) and the ACI volatility design of [Gibbs and Candès \(2021\)](#) on our data
 - ▶ extend: Romanian, Polish, Hungarian and Czech inflation, load and stock indices; covariates in context (energy prices, calendars); online CQR to 95% and VaR 1%; pooled tests with Holm and BH
 - ▶ pre-register: series, origins after 1 November 2025, horizons, losses, baselines and the decision rule; forecasts for the next months written down before the data arrive
- ▣ Deliverables follow the course rules: repository, report, AI_USE.md, AI_ERRORS.md, oral defence

Key takeaways

- ▣ Foundation models amortise estimation over a corpus; design choices (scaling, patches, quantile range) set what they can and cannot forecast
- ▣ On our data the best zero-shot model is competitive with domain models; gains depend on the task and do not survive cherry-picking
- ▣ Evaluate after the release, pool across series, correct for multiplicity
- ▣ Conformal prediction gives finite-sample marginal coverage under exchangeability; under dependence, weights, ensembles and feedback restore long-run coverage
- ▣ Conditional coverage is impossible in general: diagnose it with Christoffersen tests and coverage by regime

Self-assessment

Questions

- ▣ Why can Chronos not forecast a value above 15s?
- ▣ Why does a stationary AR(1) violate exchangeability?
- ▣ What is the law of the coverage of split conformal given the calibration set?
- ▣ Which quantity stays bounded in the proof of the ACI theorem?
- ▣ Why is a post-release window necessary in a foundation-model benchmark?

Next:

- ▣ Causal inference for time series
- ▣ Granger and structural causality, causal discovery, policy evaluation

Appendix: proof of split conformal coverage

- Condition on the training set, so $\hat{\mu}$ and s are fixed; exchangeability of $Z_1, \dots, Z_{n+1}$ implies exchangeability of $S_1, \dots, S_{n+1}$
- With no ties, the rank R of S_{n+1} among $S_1, \dots, S_{n+1}$ is uniform on $\{1, \dots, n+1\}$ (every ordering is equally likely)
- $S_{n+1} \leq S_{(k)}$ (order statistic of the first n) $\iff R \leq k$; so $\Pr\{Y_{n+1} \in \hat{C}\} = k/(n+1)$
- $k = \lceil (n+1)(1-\alpha) \rceil$ gives $1-\alpha \leq k/(n+1) < 1-\alpha + 1/(n+1)$; ties only increase coverage
- Given the calibration set, coverage is $F(S_{(k)})$ with F the score distribution; $F(S_i)$ are i.i.d. uniform, so $F(S_{(k)}) \sim \text{Beta}(k, n+1-k)$

Appendix: the two online guarantees

- **ACI:** if $\alpha_t < 0$ then $q_t = +\infty$ and $\text{err}_t = 0$, so $\alpha_{t+1} = \alpha_t + \gamma\alpha > \alpha_t$; if $\alpha_t > 1$, $\text{err}_t = 1$ and α_t falls
 - ▶ hence $\alpha_t \in [-\gamma, 1 + \gamma]$ for all t ; summing the updates,
$$\gamma \left| \sum_{t \leq T} (\alpha - \text{err}_t) \right| = |\alpha_{T+1} - \alpha_1| \leq \max\{\alpha_1, 1 - \alpha_1\} + \gamma$$
- **Quantile tracking:** if $q_t > B$ then $\text{err}_t = 0$ and q_t falls by $\eta\alpha$; if $q_t < 0$ then $\text{err}_t = 1$ and q_t rises
 - ▶ hence $q_t \in [-\eta, B + \eta]$ (starting inside); $\eta \sum_{t \leq T} (\text{err}_t - \alpha) = q_{T+1} - q_1$, so the average gap is at most $(B + \eta)/(\eta T)$
- Neither proof uses any probability: the guarantees hold for every sequence, which is why they say nothing about conditional coverage

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